

Security Marking (select from the drop-down list)

Sole Source (Including Brand Name) Justification - Simplified Procedures for Certain Commercial Items

NOTE: If a Justification was approved for the preceding acquisition, a copy of the approved Justification for the predecessor action must be included in the staff package for approval of the instant Justification. This applies to Justification staff packages that are submitted for review and approval at a level above the contracting officer. The predecessor Justification will be used as a reference document by the approving official.

Is this a new or amended J&A Document? ☒ New ☐ Amended (Prior to Award Only!)

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)? ☐ Yes ☐ No

Funding level for this acquisition: ☒ >SAT and ≤ \$750K ☐ > \$750K and ≤ \$15M

Contracting Activity: AFSC/PZIMC

Purchase Request / Local ID Number: Not Yet Assigned

Program / Project (and PE, if applicable): H6TM06 Energy dispersive micro-focusing X-ray fluorescence spectrometer (μEDXRF) with poly-capillary optics and turret

Program Type (PEO or Other Contracting): Operational

Authority: ☒ AFFARS 5313.501 - 10 U.S.C. 3572, Implementation of Simplified Acquisition Procedures (41 U.S.C. 1901)
☐ AFFARS 5313.501 - 10 U.S.C. 3403, Special Emergency Procurement Authority (41 U.S.C. 1903)

Estimated Contract Cost (including options): [REDACTED] Justification Type: ☐ Class ☒ Individual

COORDINATION ([AFFARS 5306.304\(a\)](#))

Date 19 Nov 2025	Program Manager Jeffrey D. Childs, PhD Lead Chemist 76 MXSS/MXDTAA	Signature CHILDS.JEFFREY.DALE.1 271967531 Digitally signed by CHILDS.JEFFREY.DALE.1271967531 Date: 2026.01.15 13:34:04 -06'00'
Date 20 Jan 2026	Contracting Officer Katherine Galgano AFSC PK/PZIMC	Signature GALGANO.KATHERINE. ELIZABETH.1387095779 Digitally signed by GALGANO.KATHERINE.ELIZABETH.1387095779 Date: 2026.01.20 15:10:01 -06'00'

[APPROVAL \(AFFARS 5306.304\(a\)\)](#)

Date	Chief of the Contracting Office (COCO) [Official Signature Block]	Signature
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I. Contracting Activity.

AFSC/PZIMC
3001 Staff Dr.
Tinker AFB, OK 73145

II. Nature and/or description of the action being processed.

This individual Justification and Approval (J&A) authorizes use of other full and open competition procedures to procure one MICRO-FOCUSING XRF SPECTROMETER.

The resulting contract will be a one-time, sole source, commercial, firm fixed-price (FFP), negotiated action for a commodity with incidental services. This acquisition is not a bridge action.

Contractor: Gatan, Inc.

[REDACTED]
Estimated Delivery: 7 months ARO

III. Description of supplies/services required to meet agency needs.

This acquisition is for the furnishing, shipping, assembling, installing, verification, calibration, testing, and training of firm fixed-priced contract will provide for the procurement, installation, calibration, verification and training of a commercial MICRO-FOCUSING XRF spectrometer and accessories for use in the Analytical Chemistry Lab at Tinker AFB, OK. At this time there are no projected quantities or services nor is there a potential sustainment life cycle for follow on requirements. The μ EDXRF instrument shall detect all elements from Sodium to Uranium with sufficient resolution, sensitivity, accuracy, precision, and reproducibility to satisfy all items in this SOW. Briefly, The Analytical Chemistry Section, Oklahoma City Air Logistics Center (OC-ALC), Tinker Air Force Base, OK is responsible for the chemical analysis of metals and other materials used in the United States Air Force (USAF) aircraft and jet engines managed at OC-ALC. Alloy families frequently encountered include iron base, nickel base, cobalt base, aluminum base, titanium base, magnesium base, and copper base. μ EDXRF-spectral-mapping of samples from aircraft comprised of the above alloys is required to support depot activity involving engine wear debris analysis, aircraft/engine filtrate-particle analysis, plating operations, plasma spray operations, cleaning, stripping, etc. Another type of sample is a 30mm x 30 mm filter patch that contains Thousands of particles corresponding to engine alloy classifications: M50, 9310, 4340, 17-4, 310, 347, A286, Inco 718, LA steel, Ag, W carbide, Cr plate, Ni plate, 316 steel, 304 steel, geological materials corresponding to the inorganic components of common dirt (quartz/gypsum/alkali-feldspars/ aluminosilicates), cellulose fibers, and metal-oxide-corrosion products where particles range in size over four orders of magnitude from 1 mil² to 5000 mil² with the μ EDXRF-spectral mapping being used to determine total area of all particles, individual-particle-areas, and elemental composition of each particle.

Supply or Service: Supply

[REDACTED]
Estimated Period of Performance: 7 months ARO

Estimated Delivery Date: 7 months ARO
[REDACTED]

IV. Statutory authority permitting other than full and open competition.

10 U.S.C. 2304(c)(1), as implemented by FAR 6.302-1(a)(2) - Only One Responsible Source and No

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Other Supplies or Services Will Satisfy Agency Requirements and 10 U.S.C. 2302(b) implemented by DAFFARS 5313.501 (a)(1)(ii) - Implementation of Simplified Acquisition Procedures (41 U.S.C. 1901)

V. Demonstration that the contractor's unique qualifications or nature of the acquisition requires the use of the authority cited above (applicability of authority).

The Micro-Focusing XRF Spectrometer is necessary to replace nonfunctional and obsolete equipment while increasing technological capabilities. Gatan, Inc. designs and manufactures the EDAX OrbIS II, Energy dispersive micro-focusing X-ray fluorescence spectrometer (μ EDXRF) with poly-capillary optics and turret. The distinguishing characteristics for the EDAX ORBIS II include multiple collimators for large samples, the optics being mounted on a revolving turret and both perpendicular video camera collinear with x-ray excitation from polycapillary and collimators. These distinguishing characteristics were discussed with other sources during Market Research methods as discussed in section VI. Gatan Inc. was identified as the only capable source able to meet the distinguishing characteristics provided in the Edax Orbis II Micro-Focusing XRF Spectrometer.

VI. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable.

Methods used to complete the Market Research were telephone calls, Internet searches, and consultation with SMEs. Each of these methods revealed that Gatan, Inc. is the only capable source to provide the Edax Orbis II Micro-Focusing XRF Spectrometer. Other sources provide spectrometers but they do not provide a spectrometer that meets the technical requirement of the Statement of Work (SOW).

VII. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

The Contracting Officer will ensure the items identified for procurement are negotiated at a fair and reasonable price to the Government. The contract will be awarded IAW FAR Part 12 and will utilize Simplified Acquisition Procedures.

VIII. Description of the market research conducted and the results, or a statement of the reasons market research was not conducted.

Market research was conducted through internet searches, review of other Government contracts, communications with potential vendors and known manufacturers, and careful study of the available vendor's product literature. An internet search of ANALYTICAL-INSTRUMENTATION was conducted looking for companies that could provide the MICRO-FOCUSING ENERGY DISPERSIVE X-RAY FLUORESCENCE SPECTROMETERS including a search of the American Chemical Society Division of Analytical Chemistry (<https://acsanalytical.org/pittcon>). Through market research it was determined that only one vendor could perform IAW the SOW.

IX. Any other facts supporting the use of Other Than Full and Open Competition.

None that aren't already listed above.

X. List of any sources that expressed, in writing, an interest in the acquisition.

IXRF, Inc. CAGE CODE: 04QJ1

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Business Size: Small Business

Bruker AXS LLC. CAGE CODE: 8NA10

Business Size: Large Business

Gatan, Inc. CAGE CODE: 0LP59

Business Size: Large Business

XI. A statement of the actions, if any, the agency may take to remove or overcome any barriers to competition before making subsequent acquisitions for the supplies or services required.

The Government will continue to monitor the market to determine if there are other companies that are capable of providing the equipment.

XII. Certification by the Contracting Officer.

As evidenced by my signature above, I have determined this document to be both accurate and complete to the best of my knowledge and belief.

XIII. Certification by the technical/requirements personnel.

As evidenced by my (our) signature(s) above, I (we) certify that any supporting data contained herein, which is my (our) responsibility, is both accurate and complete.